Unit 12: File Server Configuration

CONTENTS

Objectives

After studying this unit, you will be able to

- Understand the FTP protocol
- Understand the relevance and use of FTP
- How to start FTP server
- How to test FTP server
- How to use FTP client to test anonymous read access

Introduction

FTP stands for File Transfer Protocol. At its core, the file transfer protocol is a way to connect two computers to one another in the safest possible way to help transfer files between two or more points. To put it simply, it's the means by which files are securely shared between parties. FTP servers are the solutions used to facilitate file transfers across the internet. If you send files using FTP, files are either uploaded or downloaded to the FTP server. When you're uploading files, the files are transferred from a personal computer to the server. When you're downloaded files, the files are transferred from the server to your personal computer. TCP/IP (Transmission Control Protocol/Internet Protocol), or the language the internet uses to execute commands, is used to transfer files via FTP. So, the main points in consideration are:

- If you want to enable other users to download files from a location on your server's hard disk, and/or to upload files to that location, then one solution is to install an FTP server.
- You can think of an FTP server essentially as an area of disk space that is used for storing files, plus the software and configuration required to allow other users to upload and download files.

- When users want to upload or download from your FTP server, they use a program called an FTP client.

These communications between FTP server and FTP client take place using the File Transfer Protocol (FTP). FTP is a TCP protocol that is designed specifically for the transfer of files over a network, and it's one of the oldest Internet protocols still in widespread use.

Relevance of FTP

The availability of so many different FTP client programs, and the fact that many operating systems come with FTP software pre−installed, are indications of how relevant FTP still is today. FTP also is more efficient at large file transfers than HTTP and requires a password for access. FTP keeps a log of data transmission, where HTTP does not. All these components combined create a system that the common user might not notice, but which is incredibly important for professionals working with IT.FTP and its derivatives are the best choices when dealing with the manipulation of larger quantities of data. Rather than simply giving one-way access, FTP services allow users an enormous amount of control, and this can be extremely beneficial to individuals and business which update regularly.

Security of FTP

FTP is generally considered to be an insecure protocol because it relies on clear-text usernames and passwords for authentication and does not use encryption. Data sent via FTP is vulnerable to sniffing, spoofing, and brute force attacks, among other basic attack methods. It is not considered a secure protocol, because communication between the FTP client and server are unencrypted. Consequently, Secure FTP (SFTP) is also becoming popular (and, indeed, is part of the openssh package that comes with Red Hat Linux 9).It's also possible to configure your FTP server in other ways, for example by forcing users to log in, or by using access control lists (ACLs) to allow different rights to different groups of users.

Did you know?

The FTP was not built to be secure. The communication between the FTP client and server are unencrypted.

Anonymous FTP Access

- Anonymous FTP is called *anonymous* because you don't need to identify yourself before accessing files. In fact, many FTP servers still allow anonymous FTP access, which means that the FTP server allows any user to access its disk space and download its files.
- Anonymous FTP access is used mostly to enable users to access freely available documents and files via the Internet without access control.

Despite the security issues, FTP remains popular − it's fast and easy to use, and it is the Internet standard protocol for file transfer.

FTP Servers in the Red Hat Linux Distribution

There are few FTP servers available in Red Hat Linux distribution. These are:

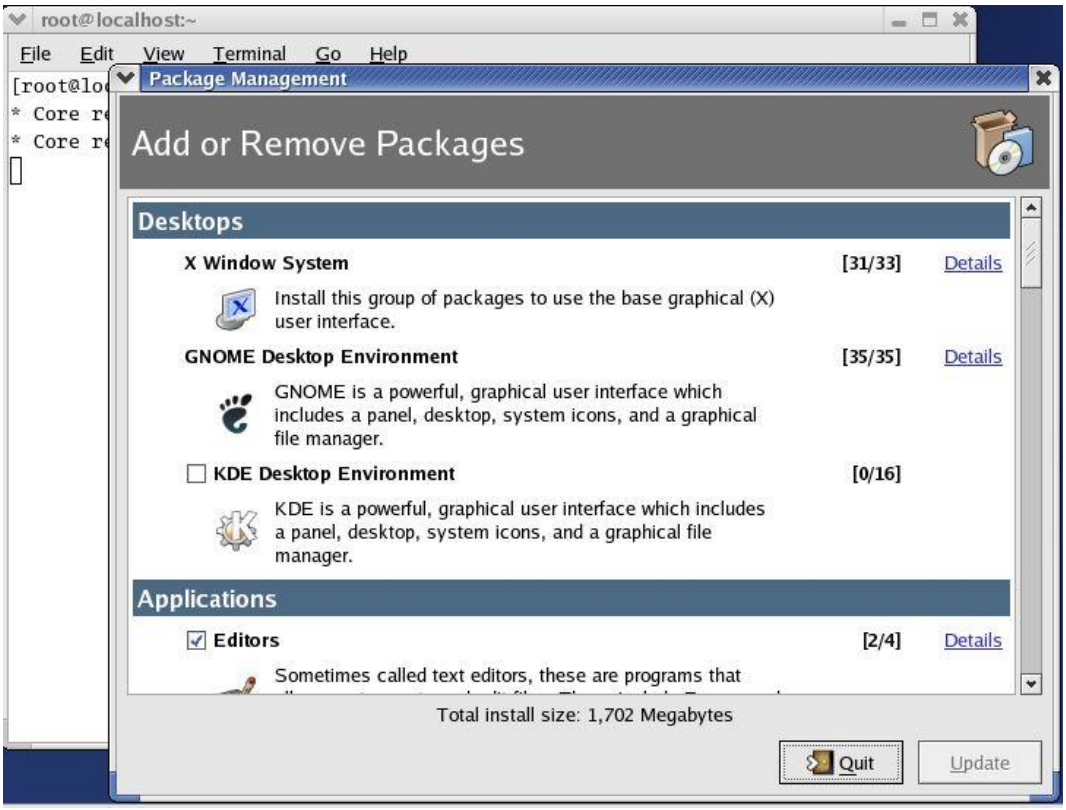
FTP Server	Remarks
vsftpd	It is a simplified FTP server implementation. It is designed to be a very secure FTP server and can also be configured to allow anonymous access.
TUX	It is a kernel−based, threaded, extremely high-performance HTTP server, which also has FTP capabilities. TUX is perhaps the best in terms of performance but offers less functionality than other FTP server software. TUX is installed by default with Red Hat Linux 9.

wu-ftpd	It is a highly configurable and full−featured FTP daemon, which was popular in earlier versions of Red Hat Linux but has since given way to the more security−conscious vsftpd.
gssftpd	It is a kerberized FTP daemon, which means that it is suitable for use with the Kerberos authentication system.

12.1 Installing the vsftpd FTP Server

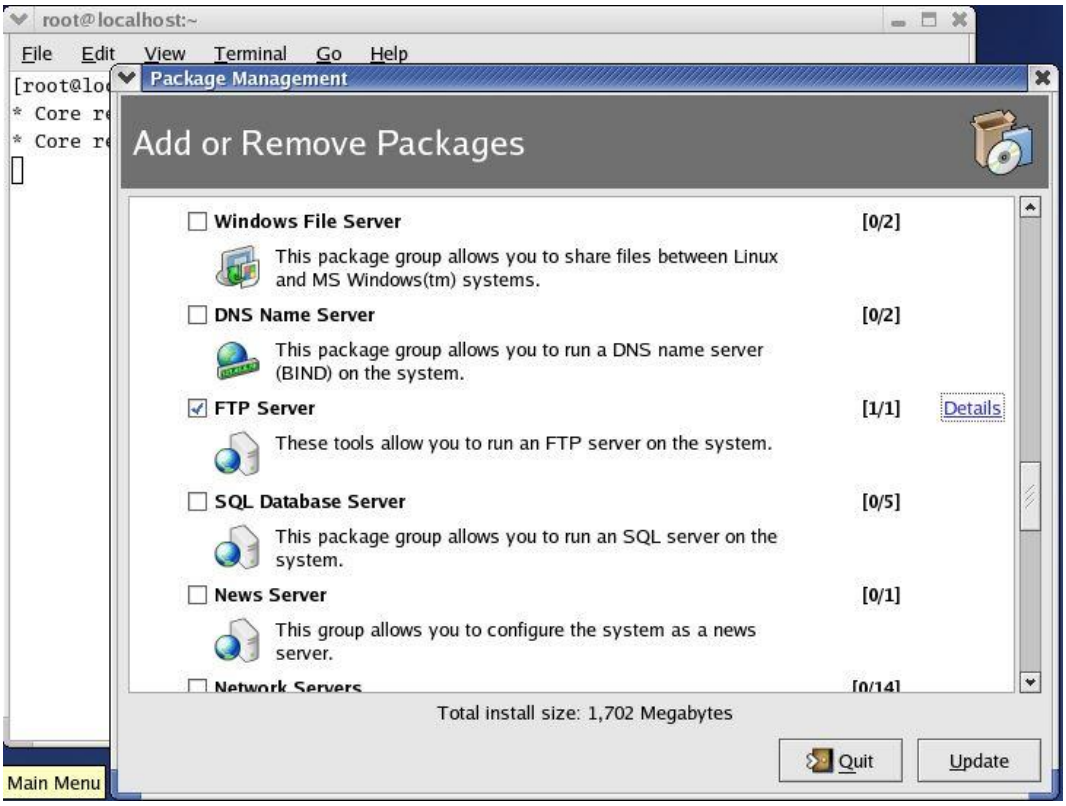
The easiest way to install the vsftpd FTP Server package is via the RPM GUI tool. There are two ways by which we can open the RPM GUI tool. These are:

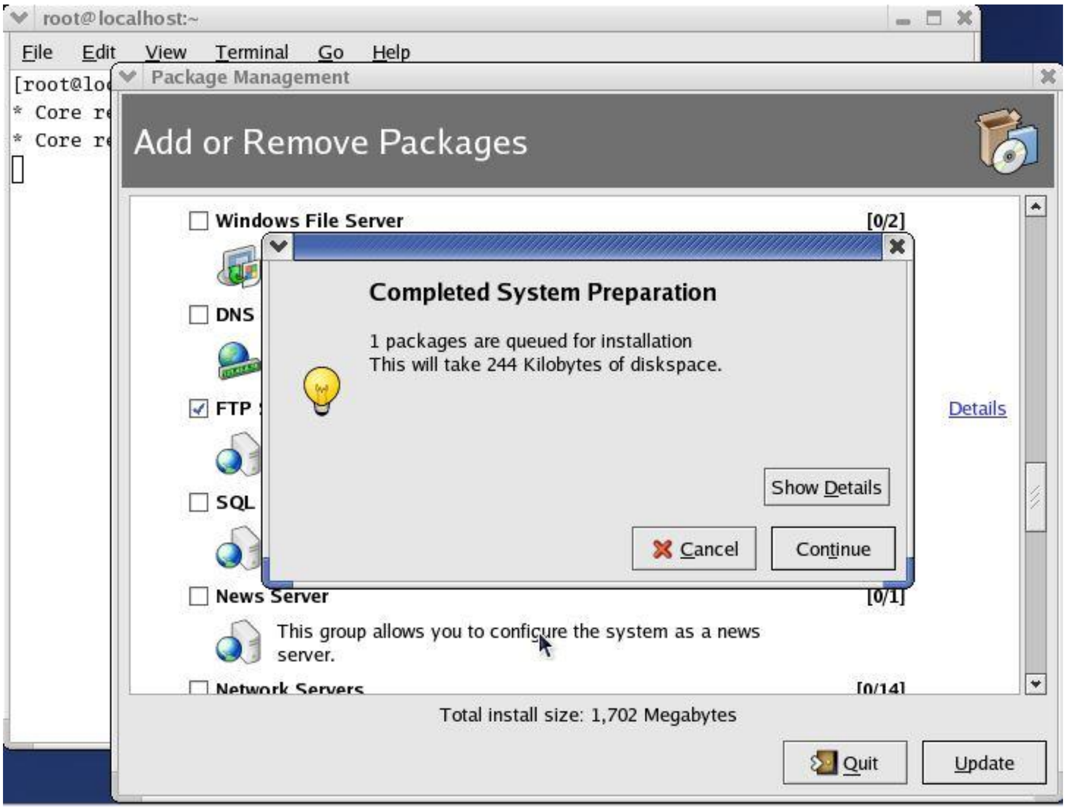
- Main Menu | System Settings | Add/Remove Applications
- $ redhat-config-packages

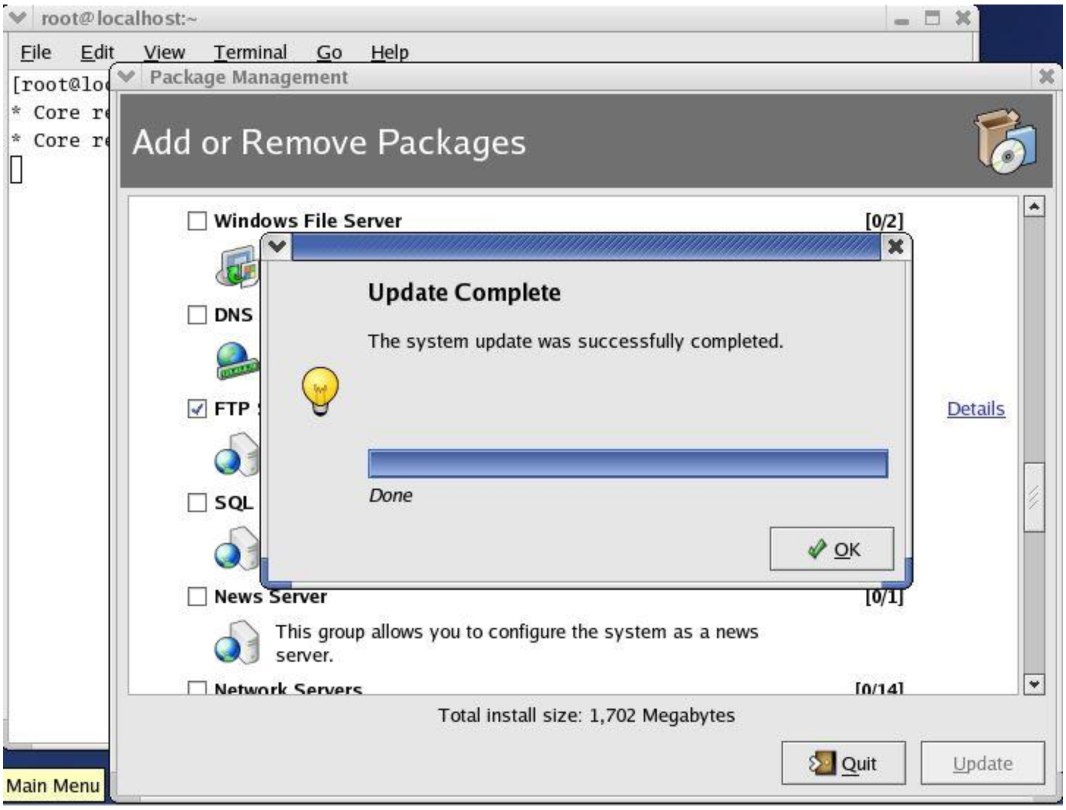


12.2 Installing the vsftpd FTP Server

For installation of FTP server, we will see the category 'Servers'. In this we will click on FTP server. By clicking on details link, we can see the packages available in that. As in FTP server we have just one package. After clicking on update button, we can


root@localhost:~
File Edit View Terminal Go Help
Package Management
Add or Remove Packages
Windows File Server [0/2]
This package group allows you to share files between Linux and MS Windows(tm) systems.
DNS Name Server [0/2]
This package group allows you to run a DNS name server (BIND) on the system.
FTP Server [1/1] Details
These tools allow you to run an FTP server on the system.
SQL Database Server [0/5]
This package group allows you to run an SQL server on the system.
News Server [0/1]
This group allows you to configure the system as a news server.
Total install size: 1,702 Megabytes
Quit
Update
Main Menu


root@localhost:~
File Edit View Terminal Go Help
Package Management
Add or Remove Packages
Windows File Server [0/2]
Completed System Preparation
1 packages are queued for installation
This will take 244 Kilobytes of diskspace.
Show Details
Cancel
Continue
Details
News Server [0/1]
This group allows you to configure the system as a news server.
Total install size: 1,702 Megabytes
Quit
Update

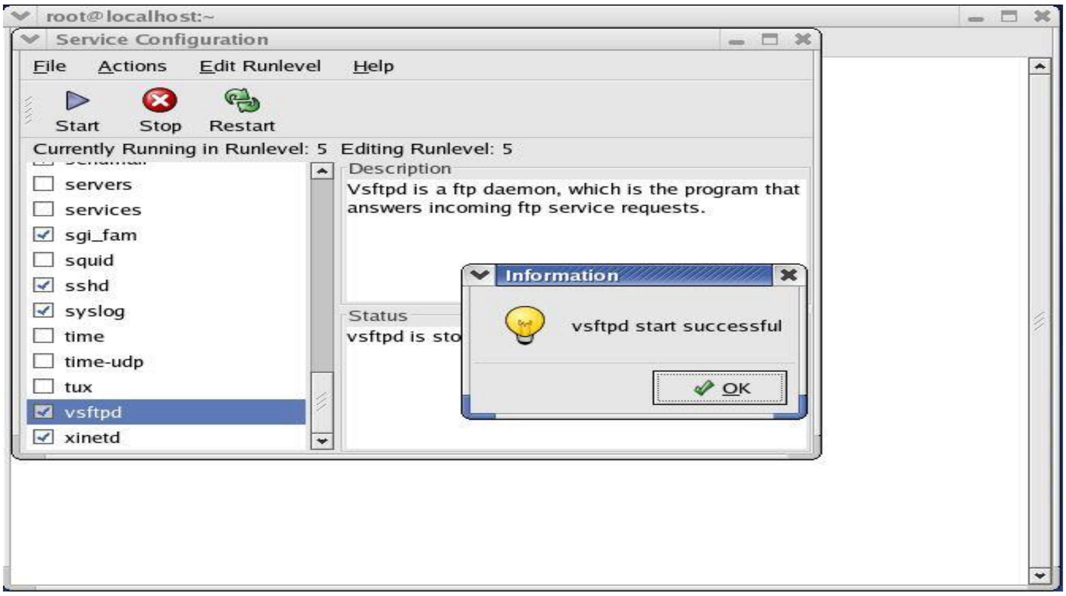


12.3 Starting your FTP Server

When the installation is done. We need to start the FTP Service, we can use the Service Configuration tool. There are again two ways to start FTP server. These are:

- Main Menu | System Settings | Server Settings | Services
- $ redhat−config−services

Again, you'll be prompted for the root password, unless you're already logged on as root. After logging in as a root, it will open service configuration dialog.



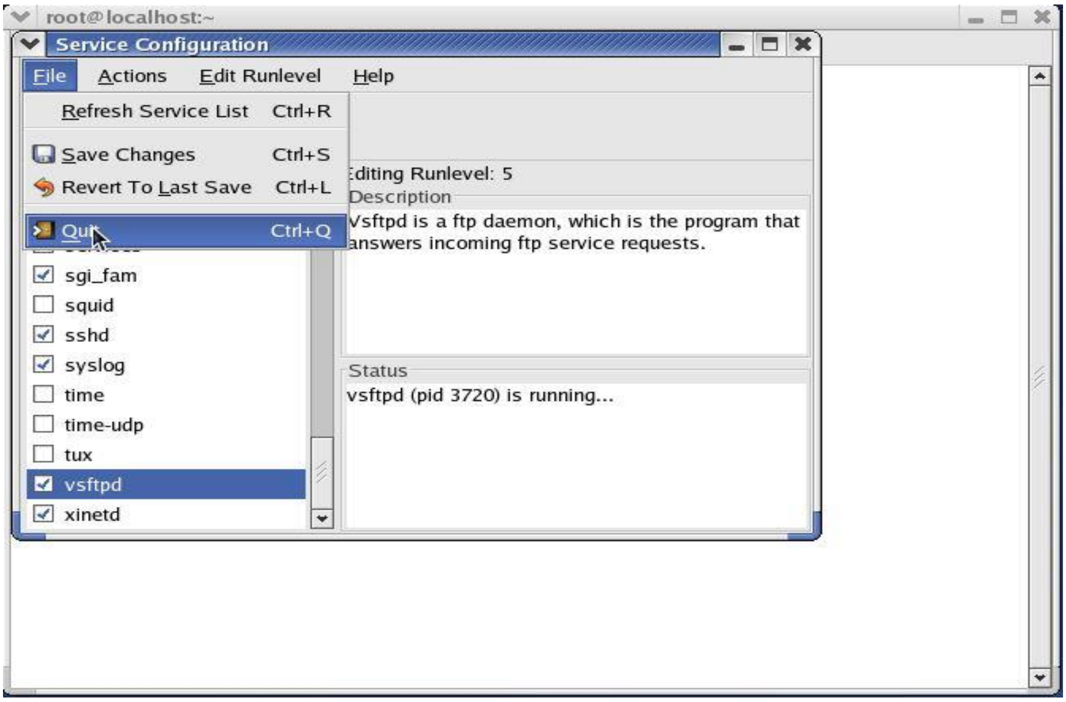
Here we need to start the service. By clicking on start button, it will start the service. After it, we need to save the changes and quit. It's also possible to start and stop these FTP services from the command line, using the service command to start and stop the vsftpd script:

```
# service vsftpd start
Starting vsftpd:                                        [ OK ]
# service vsftpd stop
Stopping vsftpd:                                        [ OK ]
```

Again, if you run the script without an option, the resulting usage message reveals all the available options:

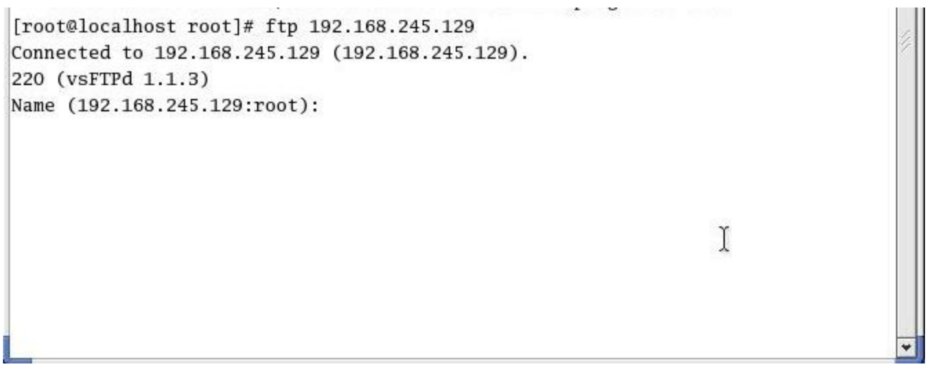
```
# service vsftpd
```

Usage: vsftpd {start|stop|restart|condrestart|status}



12.4 Testing Your FTP Server

After setting up the FTP server, we need to test it. From the client side, we will test the server whether it is working fine or not. From a command line, issue the ftp command to start an FTP session, naming your FTP server as the server that you want to connect to.You should get a Name login prompt like the one shown above − this is enough to confirm to us that the vsftpd server is running. Press Ctrl−C to terminate this FTP session and return to the command line.



Configuring an Anonymous FTP Server for File Download

Anonymous users cannot read from just any directory on your Linux server. By default, the vsftpd package creates a directory tree starting at /var/ftp, and enables 'anonymous read access' to this directory and the directory structure beneath it. For this we'll adopt the role of one of these users, and run a client FTP session to access the FTP server,examine the contents of the FTP site, and download a copy of the test file.

Setting up the FTP Server

All we need to do here is place some test content somewhere under the /var/ftp directory, so that other users can access it. The owner of the /var/ftp is the root account, and by default is the only one with permission to write to the directory. For this we need to enter as a root user, so use a command line to switch to the root user:

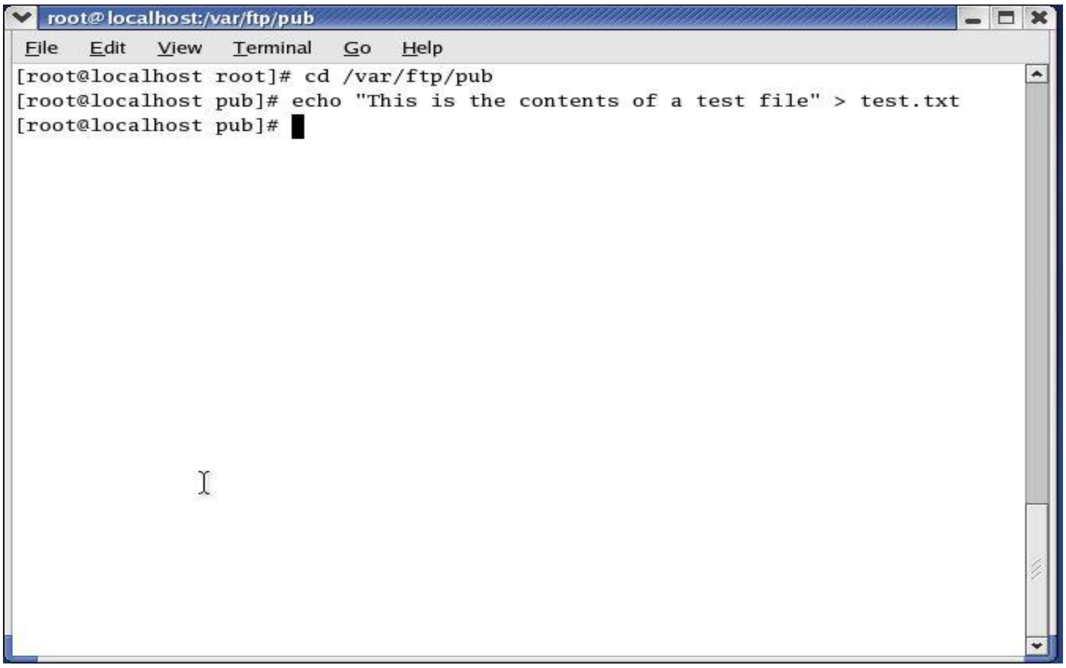
$ su −

Password:

Then you can place whatever content you want under the /var/ftp directory. For example, you can easily use acommand such as echo to create a simple test file:

cd /var/ftp/pub

echo "This is the contents of a test file!" > test.txt



12.5 Using an FTP Client to Test Anonymous Read Access

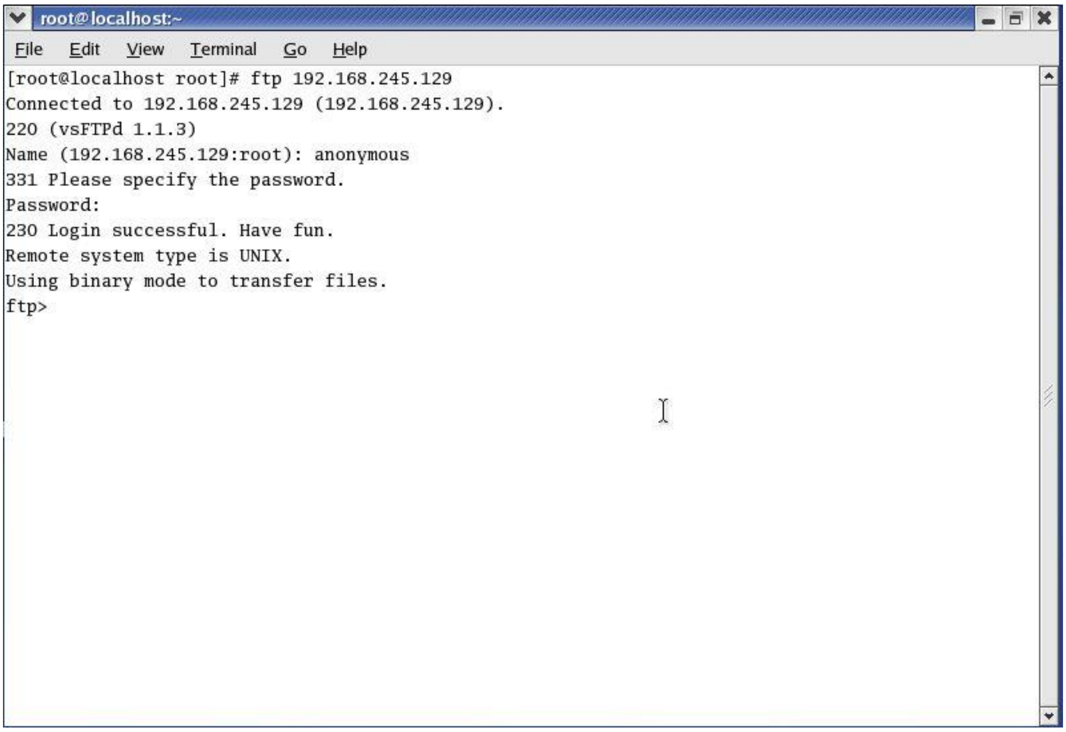
Now you can test for anonymous read access, by using an FTP client to try to grab a copy of this test file via an FTP connection. You can use any FTP client, and you can test from a Windows or Linux machine −provided the client machine can see the FTP server across a network. You can even use your Linux server as a client if you have only one machine.For example, in both Windows and Linux you can use the ftp program at the command line.In the following, we'll use the ftp program as FTP client to connect to the FTP server, examine the contents of the FTP site, and then download the file test.txt:

1) Start by connecting to the FTP server. When you're prompted for a username, specify anonymous (as shown below) or ftp to indicate that you want anonymous access:

$ ftp 192.168.245.129
Connected to 192.168.245.129 (192.168.245.129).
220 (vsFTPd 1.1.3)

Name (192.168.0.99:none): anonymous
331 Please specify the password.
Password:
230 Login successful. Have fun.
Remote system type is UNIX.

Using binary mode to transfer files.

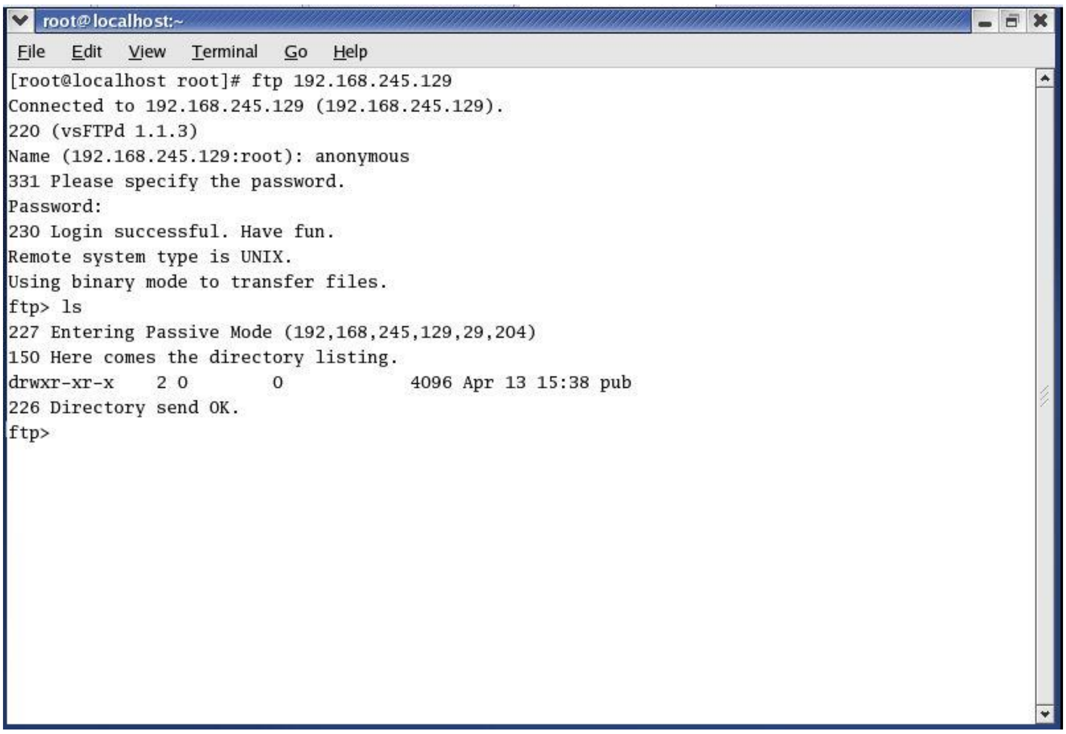


2) Now, we can start to examine the contents of the FTP site that are available to users with anonymous access. For example, here we'll use the ls command to examine the contents of the FTP root directory which happens to be the directory /var/ftp on the server:

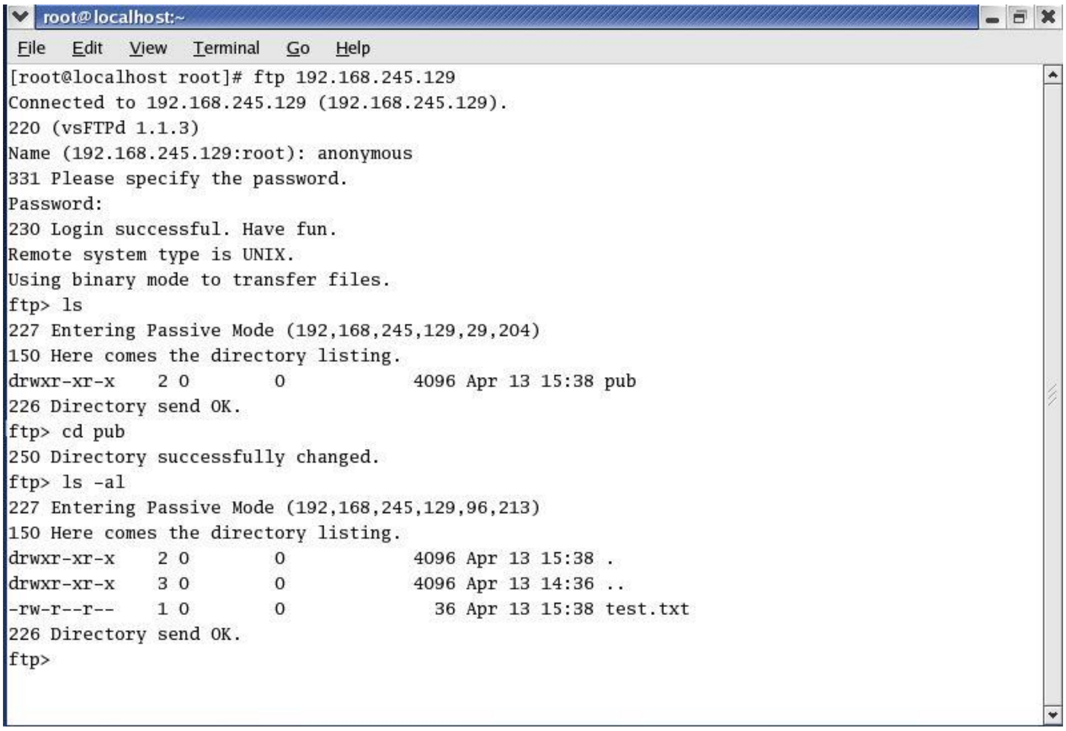
ftp>ls
220 Entering Passive Mode (192,168,245,129,29,204)
150 Here comes the directory listing.
drwxr−r−x 2 0 0 4096 Apr13 15:38 pub

226 Directory send OK.

This shows that the root directory contains just one subdirectory, called pub. Now we'll use cd to change to this directory, and we'll list its contents.



```
root@localhost:~
File  Edit  View  Terminal  Go  Help
[root@localhost root]# ftp 192.168.245.129
Connected to 192.168.245.129 (192.168.245.129).
220 (vsFTPd 1.1.3)
Name (192.168.245.129:root): anonymous
331 Please specify the password.
Password:
230 Login successful. Have fun.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,245,129,29,204)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 pub
226 Directory send OK.
ftp>
```

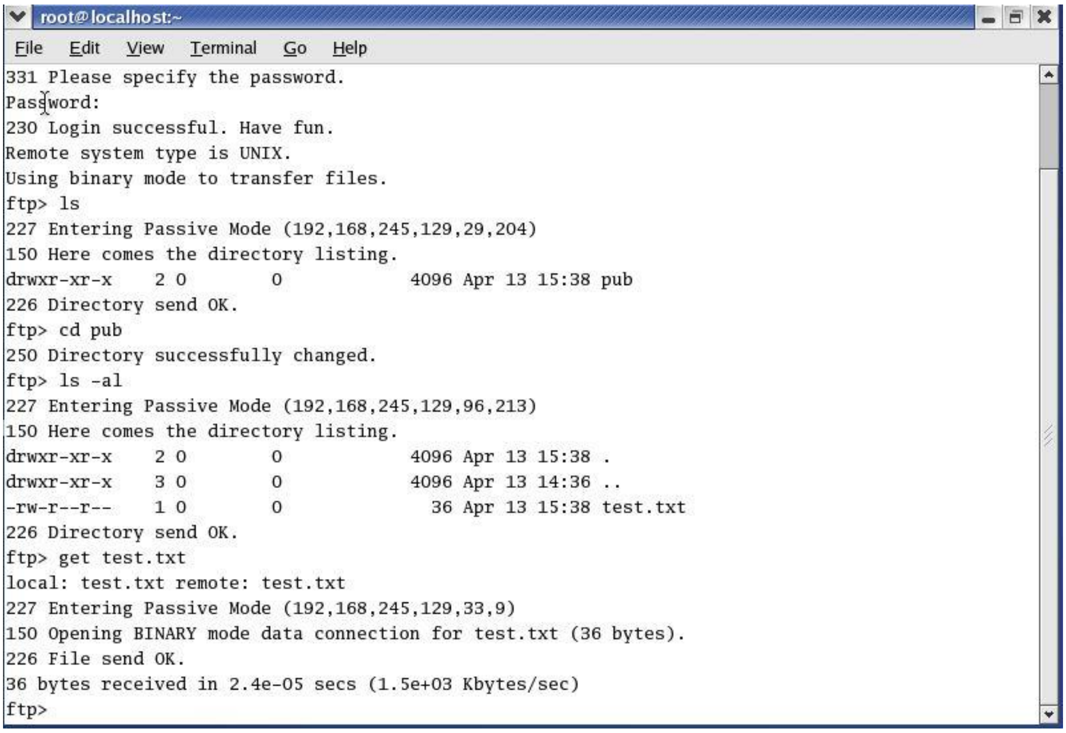


```
root@localhost:~
File  Edit  View  Terminal  Go  Help
[root@localhost root]# ftp 192.168.245.129
Connected to 192.168.245.129 (192.168.245.129).
220 (vsFTPd 1.1.3)
Name (192.168.245.129:root): anonymous
331 Please specify the password.
Password:
230 Login successful. Have fun.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,245,129,29,204)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls -al
227 Entering Passive Mode (192,168,245,129,96,213)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 .
drwxr-xr-x    3 0        0            4096 Apr 13 14:36 ..
-rw-r--r--    1 0        0              36 Apr 13 15:38 test.txt
226 Directory send OK.
ftp>
```

3) Now, we'll attempt to download the test.txt file we've just located. To do this, we'll use the get command:

ftp> get test.txt
local: test.txt remote: test.txt
227 Entering Passive Mode (192, 168, 245, 129, 33, 9)
150 Opening BINARY mode data connection for test.txt (36 bytes).

226 File send OK.

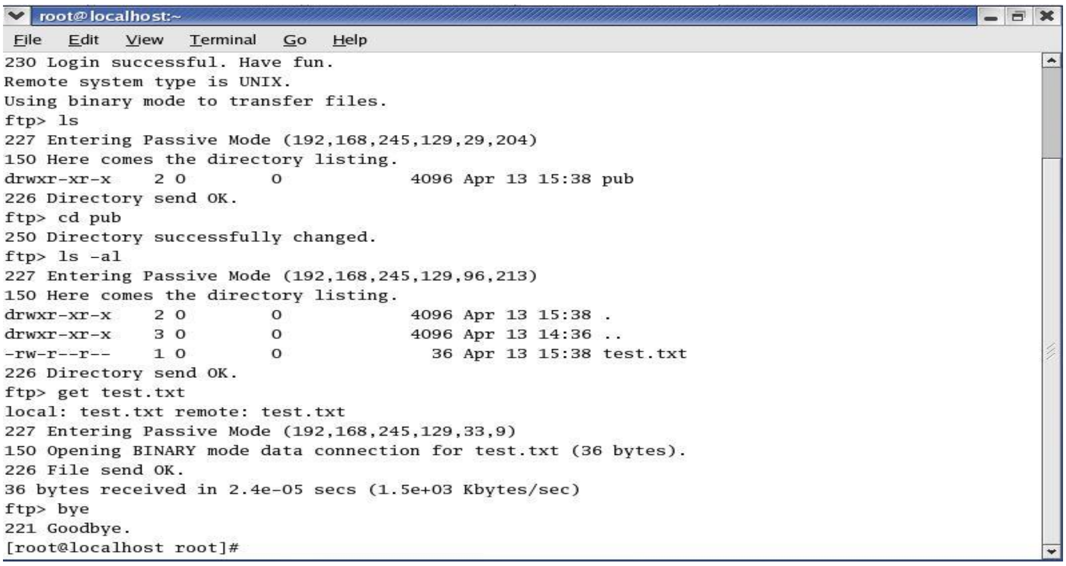


```
root@localhost:~
File Edit View Terminal Go Help
331 Please specify the password.
Password:
230 Login successful. Have fun.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,245,129,29,204)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls -al
227 Entering Passive Mode (192,168,245,129,96,213)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 .
drwxr-xr-x    3 0        0            4096 Apr 13 14:36 ..
-rw-r--r--    1 0        0              36 Apr 13 15:38 test.txt
226 Directory send OK.
ftp> get test.txt
local: test.txt remote: test.txt
227 Entering Passive Mode (192,168,245,129,33,9)
150 Opening BINARY mode data connection for test.txt (36 bytes).
226 File send OK.
36 bytes received in 2.4e-05 secs (1.5e+03 Kbytes/sec)
ftp>
```

4) Finally, we'll end the session:

ftp> bye
221 Goodbye.

$



```
root@localhost:~
File Edit View Terminal Go Help
230 Login successful. Have fun.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (192,168,245,129,29,204)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls -al
227 Entering Passive Mode (192,168,245,129,96,213)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Apr 13 15:38 .
drwxr-xr-x    3 0        0            4096 Apr 13 14:36 ..
-rw-r--r--    1 0        0              36 Apr 13 15:38 test.txt
226 Directory send OK.
ftp> get test.txt
local: test.txt remote: test.txt
227 Entering Passive Mode (192,168,245,129,33,9)
150 Opening BINARY mode data connection for test.txt (36 bytes).
226 File send OK.
36 bytes received in 2.4e-05 secs (1.5e+03 Kbytes/sec)
ftp> bye
221 Goodbye.
[root@localhost root]#
```

12.6 Configuring an Anonymous FTP Server for File Upload

Anonymous FTP users can write only to the directories that we allow them to write to.
By default, vsftpd does not allow users to upload to the FTP server at all; we must first configure the server toallow anonymous users write access to some directory. So, we'll set up the FTP server for anonymous write access first; then we'll test it again using an FTP client.

Setting up the FTP Server for Anonymous Write Access

There are four steps here. We'll need to create the folder, set the appropriate permissions, and then enable uploading in the FTP server configuration:

1. First, we need to create a writeable directory. Again, you'll need the root account for this.

   Let's createa directory called /upload (in the /var/ftp/pub directory):
   # cd /var/ftp/pub
   # mkdir upload

2. Next, we need to set the permission of the upload directory so that it allows write−only access to anonymous FTP users (so that they can write to the directory but not to download from it – this restricts file sharing among FTP users). To do this, we'll first use the chgrp command to change the group associated with the upload directory:
   # chgrp ftp upload

   Now, the owner of the folder is still root, but the directory's group is ftp − the set of FTP users. Nowwe'll use the chmod command to assign read/write/execute access to the owner, write/access only tothe group, and deny access to other users:
   # chmod −R u=rwx, g=wx, o−rxw upload

3. Finally, we must configure the vsftpd server to allow anonymous upload. To do this, we simply editthe configuration file, /etc/vsftpd/vsftpd.conf. Open this file using gedit (or your favorite text editor),and locate the following lines:
   # Uncomment this to allow the anonymous FTP user to upload files. This only
   # has an effect if the above global write enable is activated. Also, you will
   # obviously need to create a directory writable by the FTP user.
   #anon_upload_enable=YES

   Just remove the leading # character in the last line, and save the file:
   anon_upload_enable=YES

4. Finally, restart the vsftpd service by using the Restart button in the Server Configuration dialog, ortyping the following at the command line:

   # service vsftpd restart

12.7 Using an FTP Client to Test Anonymous Write Access

So, let's test our configuration with another simple session on our FTP client:

1. Connect to the client and log in (using the username anonymous or ftp) as you did before:

$ ftp 192.168.0.99
Connected to 192.168.0.99 (192.168.0.99).
220 (vsFTPd 1.1.3)

Name (192.168.0.99:none): anonymous

331 Please specify the password.
Password:
230 Login successful. Have fun.
Remote system type is UNIX.
Using binary mode to transfer files.

2. Change directory to the pub/upload directory. Try to list its contents − you'll find that you can't,

because that's the way we configured the permissions on the upload directory:
ftp> cd /pub/upload
250 Directory successfully changed.
ftp> ls
227 Entering Passive Mode (192, 168, 0, 99, 95, 148)
150 Here comes the directory listing.
226 Transfer done (but failed to open directory).

3. However, you can upload a file. To prove it, use the put command to upload a simple file like this:

```
ftp> put uploadtest.txt
local: uploadtest.txt remote: uploadtest.txt
227 Entering Passive Mode (192,168,0,99,133,229)
150 Ok to send data.
226 File receive OK.
40 bytes send in 0.000101 secs (2.1e+02 Kbytes/sec)
```

4. That's it. Now you can close the FTP session:

```
ftp> bye
221 Goodbye.
#
```

Summary

- If you want to enable other users to download files from a location on your server's hard disk, and/or to upload files to that location, then one solution is to install an FTP server.
- FTP is not considered a secure protocol, because communication between the FTP client and server are unencrypted.
- The easiest way to install the vsftpd FTP Server package is via the RPM GUI tool. One way is: Main Menu | System Settings | Add/Remove Applications and another is $ redhat-config-packages.
- Press Ctrl−C to terminate this FTP session and return to the command line.
- FTP is a TCP protocol that is designed specifically for the transfer of files over a network, and it's oneof the oldest Internet protocols still in widespread use.

Keywords

- **FTP Client:** When users want to upload or download from your FTP server, they use a program called an FTP client.
- **File Transfer Protocol:** These communications between FTP server and FTP client take place using the File Transfer Protocol (FTP).
- **vsftpd:** It is a simplified FTP server implementation. It is designed to be a very secure FTP server and can also be configured to allow anonymous access.
- **FTP Server:**FTP servers are the solutions used to facilitate file transfers across the internet. If you send files using FTP, files are either uploaded or downloaded to the FTP server.
- **TUX:** It is a kernel−based, threaded, extremely high-performance HTTP server, which also has FTP capabilities. TUX is perhaps the best in terms of performance but offers less functionality than other FTP server software. TUX is installed by default with Red Hat Linux 9.

Self Assessment

1. FTP is

A. Easy to use

B. Free

C. Internet standard protocol for file transfer

D. All of the above mentioned

2. How can we open the RPM's GUI tool?
A. Main Menu | System Settings | Add/ Remove Applications
B. $ redhat-config-packages
C. By using either of these ways
D. None of the above

3. FTP stands for
A. File transfer protocol
B. First transfer protocol
C. First temperature protocol
D. None of the above

4. Who is the owner of /var/ftp?
A. All normal users
B. Only root
C. Owner can be anyone
D. None of the above

5. Who can install FTP in the system?
A. Only root
B. Normal user
C. Any of the above
D. None of the above

6. Why FTP client program is used?
A. To upload the files to FTP server
B. To download the files from FTP server
C. Both upload and download
D. None of the above

7. Which utility is used to change the directories?
A. cd
B. changedir
C. chdirectory
D. None of the above

8. Which key combination is used to terminate the FTP session?
A. CTRL-A
B. CTRL-B
C. CTRL-C
D. CTRL-D
9. Why FTP is not considered secure?
A. Communications are fake
B. Communications are unencrypted

C. Communications are available to everyone

D. None of the above

10. FTP server can be considered as

A. Area of disk space used for storing files

B. Software to allow access

C. Configuration files to allow access

D. All of the above mentioned

11. FTP is a

A. TCP protocol

B. HTTP protocol

C. SMTP protocol

D. None of the above

12. SFTP stands for

A. Second file transfer protocol

B. Secure file transfer protocol

C. Steamed file transfer protocol

D. None of the above

13. In FTP server, the software and configuration files are required for

A. Giving the user access for download

B. Giving the user access for upload

C. Both mentioned above

D. None of the above

14. What indicates the relevance of FTP today?

A. Availability of different FTP client programs

B. Many OS come with FTP preinstalled

C. Both above mentioned

D. None of the above

15. Which of these are FTP servers?

A. vsftpd

B. TUX

C. Both of the above

D. None of the above

Answers for Self Assessment

1.	D	2.	C	3.	A	4.	B	5.	A
6.	C	7.	A	8.	C	9.	B	10.	D

11. A	12. B	13. C	14. C	15. C

Review Questions

1. What is a FTP server? Write the FTP servers available in Red Hat Linux Distribution.
2. Write the installation procedure of vsftpd FTP server. How can we start the FTP server?
3. After installation and testing of FTP server, how can we use it? Explain various issues encountered in this.
4. How can we configure anonymous FTP server for file download?
5. How can we configure anonymous FTP server for file upload?

Further Readings

Sandip Bhattacharya, Pancrazio De Mauro, Shishir Gundavaram, Mark Mamone, Kalip Sharma, Deepak Thomas and Simon Whiting, Beginning Red Hat Linux 9, Wiley Publishing, Inc.

Web Links

https://www.techopedia.com/definition/26108/ftp-server